

Abraham Ighalo

Pickering ON | 705-206-8142 | LinkedIn | aighalo@algomau.ca | GitHub | Party Rock-Projects

EDUCATION Bachelor of Computer Science

Algoma University (Sept 2020 - June 2025)

Relevant Coursework: Data Structure and Algorithms, Databases, Discrete Mathematics, Software Engineering, Analysis of Algorithms, Networking, Web Design.

Thesis: Quantum Enhanced Data Analytics for Crime Prediction (presented at 2025 SNPD International Conference)

EXPERIENCE

TD Bank | Software Engineer | Python, SQL, Dynatrace, Splunk, Power BI, GitHub, CI/CD, Rundeck (November 2025 - Present)

- Built a Python XML parser to consolidate missing columns from ServiceNow into the on-prem database, improving data integrity and reducing manual reconciliation efforts by 50%.
- Designed and implemented executive-level metrics for incident, change, and problem management using Power BI DAX measures, consolidating monitoring data from SQL to provide stakeholders with actionable insights on application performance and process health.
- Built governance dashboards for incident ticket management, enhancing ticket hygiene and improving data quality.
- Built and scheduled a data pipeline using Rundeck to pull PagerDuty data, enabling real-time dashboards for operational reporting and improving visibility into system alerts and maintenance activity.

Kyanon Digital | Poseidon Team | Data Engineering Intern | Python, Spark, Dask, MLFlow, Scikit Learn, Streamlit (Jan 2025 - June 2025)

- Implemented a recommendation model to ensure equitable personalization while maintaining user relevance and engagement.
- Designed anomaly detection models to uncover irregularities in credit card loyalty and rewards programs, supporting fraud prevention.

OLG | Performance Analyst Intern | SQL-Server, SQL, Power BI, Excel, DAX, Qualtrics (Sept 2024 - Dec 2024)

In this role, I built and maintained operational reports for the purpose of enhancing the experience of users of OLG's gaming and lottery products

- Created a custom Excel analytics report for SLT, consolidating lottery performance metrics and company health indicators for data-driven decisions.
- Restructured project portfolio reports, improving cross-departmental communication and SLT's ability to prioritize future projects.

Sault Area Hospital | Junior Data Analytics intern | SQL, Power BI, SSRS, Excel, DAX (July 2024 - Sept 2024)

- At the hospital, I performed Data Migration and Report Building. I managed the migration of 12 Operational Legacy SSRS reports to power BI.
- In this role, I built reports that were used by medical professionals to help make the best decisions for patients and staff. Among the sectors I reported include appointments, bed allocation, hazards identification and prevention.
- I also managed daily reports that were viewed by medical professionals to provide up-to-date summarised assessments of patient status, significantly enhancing speed in decision-making, operational efficiency as well as supported critical healthcare operations.

TD bank | Software Engineer intern | MYSQL, SQL, Python, Apache Spark, Data Factory, Databricks, Bitbucket, Git (May 2023 - Dec 2023)

I worked with the Fraud Engineering team to onboard customer datasets from third-party vendors and TD's credit card platforms to Azure cloud, enabling advanced fraud detection analytics targeting money mule accounts. I documented the complete onboarding framework in Confluence, creating a reusable process for future dataset integrations.

- Designed a reusable Spark-based system, reducing manual effort by 70% with JSON configurations.
- Optimized Spark RDDs, cutting 30GB file onboarding time from 3 hours to 1 hour (67% faster).
- Automated pipelines with Data Factory, eliminating 70% of manual tasks and preventing reporting delays.

PROJECTS

Product Recommendation System | Python, Pandas, NumPy, Spark, Sentence Transformers [Github](#)

- Built scalable content-based recommendation engine using PySpark, MiniLM L6 v2, PCA, and KMeans clustering
- Implemented data preprocessing and feature engineering pipelines for improved recommendation accuracy and product discoverability

Loyalty Program Fraud Detection System | Python, Pandas, Machine Learning [Github](#)

- Developed comprehensive anomaly detection system using 7 ML algorithms (supervised & unsupervised), achieving 98.6% ROC AUC with XGBoost
- Performed exploratory data analysis on 284K+ transactions, identifying key fraud patterns such as peak activity at 9PM and higher afternoon fraud rates.
- Engineered features from 284K+ transactions including reward points simulation and time-based patterns

Alke Club | JavaScript, Python, Django [Github](#)

Designed to support a new club at Algoma University, created to foster community among African students and help newcomers adapt to the Canadian culture.

- Developed a full-stack Django web application for handling user feedback and contact form submissions with front-end and back-end integration.
- Enabled secure and efficient data handling using Django's CSRF protection and built-in form validation.
- Processed user input on submission and returned real-time confirmation messages via email to enhance user experience.